

Are the Skilled Trades Safe From AI?

The already-in-it guide — the full scores by track, the six factors behind them, and what to do from where you already are.

2.5

AI EXPOSURE · ELECTRICIAN — SERVICE & MAINTENANCE
where 10 is most at risk

For anyone already in it — in the degree, or job-hunting.



Before you read this

The choice is already made — you're in an apprenticeship or training, or qualified and into the job hunt. So this guide skips the "should you go into a trade" question and goes straight to the one that's actually live: given where you already are, what do you do about it?

It's written to you directly — whether you are the one in it, or a parent reading on their behalf. Where it says "you," a parent can simply read "them"; the moves are the same either way.

What this is. A facts document, not a pep talk. It sets out what's known about where the trades sit in an AI economy, where the evidence is contested, and where we might be wrong — and then what to actually do from here.

The good news, up front. You're already in the most protected career we measure — a service electrician scores 2.5, among the lowest scores in this entire index. But "trade" isn't one destination, it's a spread from 2.5 to 5.0, and where you land inside it is decided mostly by moves still in front of you. The second half of this guide is those moves — because even here, the trade and the setting you steer toward matter.

And one thing specific to the trades — honest on both sides: this is not a consolation prize, and it is also not for everyone. The physical demands over thirty years are real, the income ceiling without business ownership is real, and someone who dislikes being outdoors in February will dislike it for a very long time. If you're in it and having second thoughts about the body or the ceiling, those are worth facing squarely — and steering around — rather than ignoring.

The short answer

For the right person, this is the strongest bet on the board — and you're already on it.

A service electrician scores 2.5 out of 10 — among the lowest scores in this entire index. No tuition debt. An on-ramp that is widening while most degree routes narrow. And the demand picture is the strongest we measure anywhere.

The protection was never "working with your hands." It was working with your hands somewhere unpredictable. That distinction is the whole guide, and it explains why the same

skills on a production line score twice as high — which is exactly why *which version you steer toward* still matters, even in the most protected career we measure.

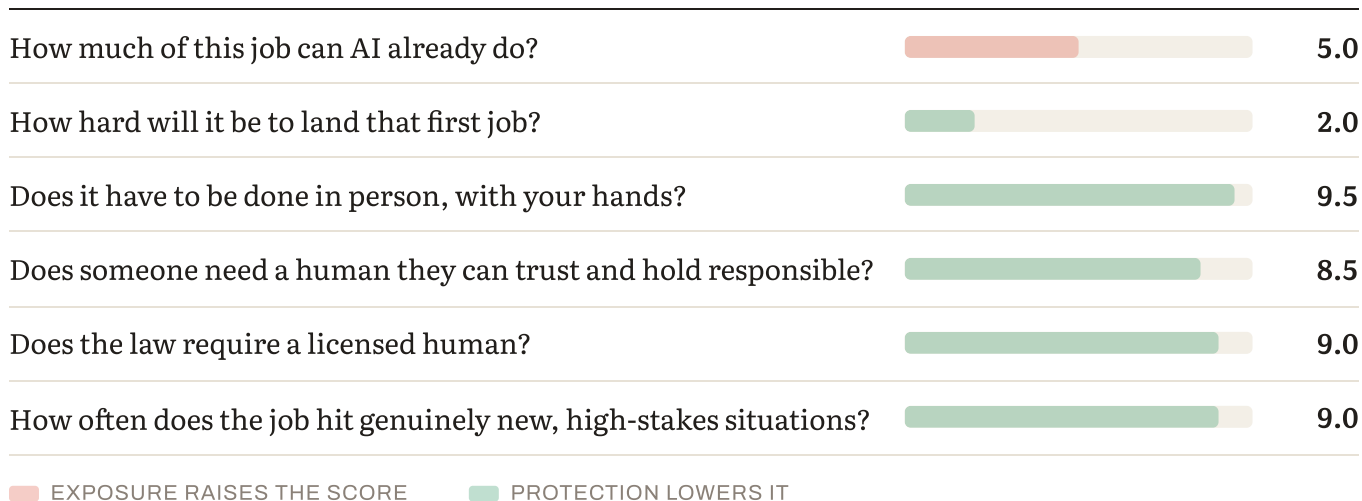
The scores

TRADE AND SETTING	2023	2025	FALL 2026	3-YR MOVEMENT	BAND
Electrician — service & maintenance	2.1	2.3	2.5	+0.4	Very low
Plumbing / HVAC — service	2.2	2.4	2.6	+0.4	Very low
Welding & fabrication — site work	2.8	3.0	3.2	+0.4	Low
Construction site trades	2.8	3.1	3.4	+0.6	Low
Production line / prefab / modular	4.0	4.5	5.0	+1.0	Moderate

Scores run 1–10, where 10 is most at risk. For context: the median profession in this edition sits around 5.5, bedside nursing scores 2.8, licensed engineering 4.0, a business analyst 7.6, and entry-level software development 8.1.

Read the table this way: the service electrician's +0.4 over three years is the slowest movement in the entire index. Entry-level software moved +1.8 over the same period. And note the last row — the same trade skills in a factory setting move at more than twice the rate of the same skills in the field. Your job, from here, is to steer toward the field end of that table, not the factory end.

Why the trades score so low — the six factors



Here's how a service electrician answers them.

HOW MUCH OF THIS JOB CAN AI ALREADY DO? (35% OF THE SCORE)

The paperwork, and essentially nothing else.

Quoting and estimating. Scheduling and dispatch. Invoicing and job admin. Code and regulation lookup. Parts ordering and inventory.

For a self-employed tradesperson, almost all of that is a benefit rather than a threat. The office burden of running a trade business is the part being automated.

What resists: diagnosing a fault in a sixty-year-old building, running new service through a wall with no drawings, working in an attic or a crawlspace or a trench, and judging whether conditions are safe to continue.

HOW HARD WILL IT BE TO LAND THAT FIRST JOB? (15%)

Among the best on-ramps we measure anywhere, and it is getting better rather than worse.

Apprenticeships are paid rather than charged for, legally embedded in licensing, and chronically undersubscribed. The route in is widening at exactly the moment most degree routes are narrowing.

BLS projects **649,300 annual job openings** in construction and extraction occupations over 2024–2034. Electricians specifically show **9% growth** — classified as "much faster than average" — with roughly **81,000 openings each year** against total employment of about 818,700.

One fork worth seeing even here, where entry is easy. The setting you land in shapes the protection you inherit. The same trade skills on a production line or in a prefab facility score 5.0

— double the field electrician's 2.5 — because that is the controlled environment automation handles best. The way through is to aim at the work that leads to a license and breadth of field service — maintenance and retrofit — rather than whichever placement is hiring fastest. An easy start in repetitive new-build or line work is not the same as landing in the most protected version of the trade.

DOES IT HAVE TO BE DONE IN PERSON, WITH YOUR HANDS? (15%)

Completely, and in the hardest possible setting for automation.

This is the factor that does most of the work, and the section on why "manual work is safe" is the wrong idea explains why "physical" is not one thing.

DOES SOMEONE NEED A HUMAN THEY CAN TRUST AND HOLD RESPONSIBLE? (15%)

Yes, and more than people assume.

Someone is being let into a home. Someone is signing off that the work is safe. When an installation fails and causes a fire or a flood, a licensed human is answerable.

DOES THE LAW REQUIRE A LICENSED HUMAN? (10%)

For electrical and gas work, comprehensively.

Licensing, inspection and certification requirements are written into building codes and enforced by inspectors. Nobody signs off their own unlicensed electrical work.

HOW OFTEN DOES THE JOB HIT GENUINELY NEW, HIGH-STAKES SITUATIONS? (10%)

Every single job.

A different building, unlabeled wiring, work someone else did badly in 1974, and a customer describing the problem inaccurately. The environment is different every time, which is precisely the condition robotics handles worst.

"Manual work is safe" is the wrong idea

This deserves its own section, because the popular version of this argument is wrong in a way that matters — and the trades are where it breaks most visibly.

The usual framing is binary: desk jobs are exposed, manual jobs are safe.

What actually matters is not whether the work is physical. It's whether the environment is predictable.

Robotics performs well in controlled settings — known dimensions, repeated tasks, fixed tolerances, no surprises. That is a warehouse, an assembly line, a modular construction facility. It performs badly in uncontrolled ones, and that gap is not closing quickly.

A service electrician spends the day in unpredictable environments by definition. That rates **9.5** on our physical factor.

The same trade skills applied to a production line rate **8.0** — still physical, but in exactly the setting robotics handles best. That single difference, plus weaker licensing and lower trust requirements, is what separates a 2.5 from a 5.0.

So the useful question for any hands-on career is not "is this manual work?" but "would two days of this job ever look the same?" If the answer is no, the protection is real. If the answer is yes, physical work offers much less shelter than people assume.

This is also the thing to watch over a career. Modular and offsite construction is growing, and it works by moving site work into factories. It is the one development that could genuinely move these scores — not because robots got better at crawlspaces, but because the industry moved the work somewhere robots already function.

What the trend shows

Service electrician: 2.1 → 2.3 → 2.5. Up 0.4 in three years — tied for the slowest movement in the entire index, and all of it administrative.

Production line / prefab: 4.0 → 4.5 → 5.0. Up 1.0, more than twice the pace.

The gap widened from 1.9 points to 2.5. The pattern is now familiar: the unpredictable end holds while the structured end drifts. What is unusual here is how little the protected end has moved at all.

How durable is the protection?

PROTECTION	STRENGTH TODAY	DURABILITY	WHERE THE PRESSURE IS COMING FROM
Physical / embodied	Strongest in the index	The one to watch	Not from robots reaching crawlspaces, but from prefabrication moving work into factories.
Regulatory / licensing	Strong for electrical and gas	Most durable	Building codes and inspection regimes move at the speed of legislation.
Judgment under novelty	Very strong	Durable	Every building is different. No two service calls are alike.
Human trust / accountability	Strong	Durable	Someone is in a home, and someone signs off on safety.

The honest read. You'll work for roughly forty-five years — longer than most graduates, because you start earlier. Over that span, the licensing and novelty protections should hold comfortably. The physical protection is the strongest today and the one with the most credible long-term threat — and the threat is industrial reorganisation rather than robotics. Which is why the direction you steer — toward field service, away from the factory — matters more than where you happen to start.

How the role will actually change

WAS THE JOB	IS BECOMING THE JOB
Working out the quote by hand	Software quotes it; you check it's right
Ringling round for parts	Ordering is automated
Paperwork in the evening	Largely gone
Diagnosing from experience	Diagnosing from experience, plus better instruments
Learning codes by memory	Looking them up instantly on site
The actual work	Unchanged

The last row is the point. In most professions the automatable layer and the skilled layer are tangled together. In the trades they are cleanly separable, and what's automating is the part nobody enjoys.

Human strengths that will matter most

1. **Diagnosis in unfamiliar conditions** — working out what's wrong in a building nobody documented. The core skill and the least automatable thing here. 2. **Physical judgment** — knowing whether something is safe to proceed with, by feel and by eye. 3. **Being trusted in someone's home** — a genuine commercial asset, and the foundation of a business. 4. **Customer handling** — explaining a problem plainly to someone worried about the cost. 5. **Business sense** — the thing that separates a \$60,000 tradesperson from a \$200,000 one.

And underneath all five, two traits the score can't grade: **curiosity and adaptability**. The specific tools — the diagnostic instruments, the thermal camera, the quoting and scheduling software — will turn over many times across a career, and so will the specializations that pay best. The tradespeople who compound are the ones who stay genuinely curious and re-tool fastest — moving toward data centers, EV infrastructure, solar and heat pumps as the demand shifts — rather than defending the one way of working they trained on. In a trade where the ceiling is reached by where you take the skill, that flexibility is itself a form of protection.

Other things to consider

What's genuinely good about it

The financial position is better than most people realize, and the comparison usually never gets made.

- **No tuition debt.** Apprentices are paid from day one — typically starting at \$15–20 an hour and stepping up annually.
- **Four to five years to journeyman status**, earning throughout.
- **BLS median wages:** electricians **\$62,350**, plumbers **\$62,970**, with top earners in both exceeding **\$106,000**.
- **Union total compensation runs roughly 60% higher** than non-union once health, pension and benefits are counted, per BLS.

- **Construction wages grew 4.2% year-over-year** into mid-2025, outpacing the national average across all occupations.

The comparison worth running: a graduate finishes at 21 or 22 with debt and enters a market where 41.5% of recent graduates are underemployed. An apprentice finishes at 22 or 23 with no debt, four years of earnings behind them, a license, and — currently — employers competing for them.

And what tradespeople report as rewarding is consistent and distinctive: the problem is solved and visible by the end of the day, the work is genuinely varied, autonomy is high, and running your own business is a normal outcome rather than an exotic one.

What's genuinely hard about it

The physical toll over decades is the real cost, and it is the thing most likely to end a career early. Knees, backs and shoulders. The work is done in heat, cold, confined spaces and awkward positions.

To be fair to the data: the nonfatal injury rate for specialty trade contractors is **2.3 per 100 full-time workers** and has been declining, driven by sustained focus on falls, struck-by, electrocution and caught-in hazards. It is safer than it was. It is still harder on a body than any desk job, and that compounds over thirty years.

The income ceiling without ownership is real. Median pay is solid and the top end is excellent, but the path to the top end usually runs through running a business — which is a different skill from the trade itself, and not everyone wants it.

It is cyclical. Construction rises and falls with credit conditions and public spending in a way that nursing and teaching do not.

And the status question is real, even if it shouldn't be. Some people find the condescension wearing, particularly from people who assume it was a fallback.

Who this work suits — and who it doesn't

The trades tend to suit people who:

- **Want the problem solved by the end of the day.** The single most-cited satisfaction.
- **Would rather be moving than sitting**
- **Like diagnosis** — the puzzle of working out what's actually wrong
- **Are practical with people** — most trades involve customers, not just materials

- **Are interested in eventually running something**

It's harder for people who:

- Genuinely **dislike physical discomfort**. Being outdoors in February is not a phase.
- Want **predictable indoor conditions**
- Need **structured career progression** with titles and ladders
- Are **conflict-averse about money** — chasing invoices is part of self-employment

If some of that second list sounds like you, it isn't necessarily a reason to leave — but it's worth facing honestly and steering around, which is what the rest of this does.

What else is moving this market

The shortage is severe and well documented. The construction industry needs to attract hundreds of thousands of net new workers in 2026 alone, against an ageing existing workforce.

And here is the detail that reframes this entire guide. Microsoft's president Brad Smith has publicly identified the electrician shortage as the single biggest constraint on data center expansion — not chip supply, not land, not permitting. Industry estimates put the requirement at 300,000-plus additional electricians just to meet AI-related data center demand. Electricians working on data center projects report earnings well into six figures.

Capital has noticed: BlackRock launched a \$100 million initiative in March 2026 aimed at training 50,000 skilled trade workers, and the US Department of Labor announced \$145 million in apprenticeship grants targeting shipbuilding, defense, semiconductors and energy.

So the AI buildout is simultaneously the thing threatening white-collar entry work and the thing driving demand for electricians. That is not irony — it is the most concrete illustration in this index of where the protection actually lies.

One honest caution. A meaningful share of current construction demand is project-driven — data centers, infrastructure programs — and those end when the builds do. The demographic case, an ageing workforce retiring faster than replacements arrive, holds regardless of the cycle. **Weight the demographics more heavily than the boom.**

The AI-native advantage — what it actually looks like

The trades are unusual here: AI helps a tradesperson mainly by removing the parts of the job that aren't the trade. Being already in it, close to real work, is exactly where that advantage gets

built.

1. Run the business side properly. Quoting, scheduling, invoicing and customer follow-up are now largely automatable. The tradespeople who use this well aren't better at the trade — they're better at running a trade business, which is where the income ceiling actually sits.

2. Use diagnostic tools without deferring to them. Thermal imaging, circuit analysis and fault-finding tools are improving. They inform judgment; they don't replace it.

3. Look things up rather than memorise them. Codes change constantly. Instant lookup is a genuine efficiency gain.

4. Specialize deliberately. Data centers, EV infrastructure, solar and battery, heat pumps. These are where the premium is and where demand is growing fastest.

5. Get to ownership. Everything in this guide points at it. The trade gives you a protected skill; the business is what converts it into a ceiling.

All five are available to you now — inside your current apprenticeship or first job, not after some future qualification. Turning them into concrete moves is the rest of this guide.

Where this can lead later

The ceiling is higher than most people assume, and it is reached differently.

- **Journeyman → master → contractor** with your own license
- **Business ownership** — the main route to the top end
- **Site supervision and construction management** — trade-to-supervision remains a fully respected path, and construction management scores 3.3
- **Specialist and high-demand work** — data centers, industrial, marine, controls
- **Inspection, code enforcement and safety**
- **Teaching the trade** — instructor roles at colleges and apprenticeship providers
- **Estimating and project management** — note this moves toward the desk, and the score climbs accordingly

The pattern holds here as everywhere: moving away from the physical work raises exposure. Estimating scores 6.3. It's a good career and a different one. That pattern is the compass for every choice below.

Repositioning from here

The trade-selection stage is behind you. The useful questions now are different — and the good news bears repeating: you're already in the most protected career on the board, and where you land inside the 2.5-to-5.0 spread is decided mostly by things still in front of you. In a strongly protected trade, repositioning isn't about escaping exposure — it's about leveling up: aiming at the most durable version of the work, and at the ceiling.

If you're still in the apprenticeship or training

Three levers, in order of leverage.

- 1. The field, not the factory, is the whole game.** Service, maintenance and retrofit work is the protected end — service electrician 2.5, plumbing/HVAC service 2.6 — while production-line, prefab and modular work scores double (5.0) because it's the controlled setting automation handles best. If your placement is drifting toward repetitive line or new-build work, that's the single highest-leverage thing to correct while you still can. Aim at the work where the environment is different every time.
- 2. The license and breadth, while you're building them.** The license is the legal protection, and breadth of work — domestic, commercial and industrial — determines the ceiling. Confirm your program leads cleanly to the license rather than leaving gaps, and push for the widest range of work you can get. That breadth is far cheaper to acquire during the apprenticeship than to bolt on later.
- 3. A high-demand specialism, and the trade itself.** Electrical leads on pay, growth and protection, and sits directly in front of the AI-driven data center demand wave. If you're mid-training and the trade is still movable, that matters. Within any trade, steer toward the premium specialisms — data centers, EV infrastructure, solar and battery, heat pumps — because that's where demand grows fastest and the ceiling is highest.

If you're qualified and you're job-hunting

Here the setting on the offer matters more than the wage, because the exposed end of this trade pays too.

- **Repetitive production-line, prefab and modular work is the 5.0.** It's available in the shortage and it's the setting the score says drifts fastest. Taking one isn't fatal — but take it as a deliberate stepping-stone with an exit in mind, not as the plan.

- **Aim at field service, maintenance and retrofit (2.5–2.6).** More protected, closer to the unpredictable environment automation handles worst, and where the ageing building stock guarantees a large and growing share of work that cannot be prefabricated.
- **Chase breadth and the license.** A seat that gives you commercial and industrial work and counts toward the license beats a narrow domestic-only role that caps the ceiling — even at a smaller employer, where breadth and responsibility tend to arrive sooner.
- **Lean into the high-demand specialisms.** Data center, EV, solar and battery and heat-pump work is where the premium is and where earnings run well into six figures.

The AI-native move, from where you stand

You can't change how far the tools have come, but you can be the tradesperson who directs them rather than ignores them. The five capabilities above — run the business side, use diagnostic tools without deferring, look things up, specialize, get to ownership — are all available right now, inside a current apprenticeship or first job. The most powerful for someone already in it is the first paired with the last: **get genuinely good at running a trade business, because that's where the income ceiling actually sits.** The trade gives you a protected skill; the automated quoting, scheduling and invoicing tools make the office side of ownership manageable in a way it never was. You can start building that this year, not after some future qualification.

The career-hunting checklist

The analysis tells you where the exposure is. This is what to do about it — the part that's actually in your hands. Most of the ceiling in the trades is built here, in these choices and habits, rather than handed to you. The first list is for anyone still in the apprenticeship or training; the second is for the job hunt itself. In the most protected career on the board, the work here is less about escaping exposure and more about aiming at the most durable version of the trade.

During the apprenticeship — build toward the license and the ceiling

- **Lean electrical, or confirm your trade on purpose.** Electrical leads on pay, growth and protection (service electrician 2.5), and it sits directly in front of the AI-driven data center demand wave. Plumbing/HVAC service (2.6) is close behind. If the trade is still movable, move it deliberately.
- **Aim at the field, not the factory.** Service, maintenance and retrofit work is the protected end (2.5–2.6); production-line, prefab and modular work scores double (5.0) because it's the controlled setting automation handles best. Get your experience where the environment is different every time.

- **Treat the license as the product, not the placement.** Confirm the program leads cleanly to the license and to breadth of work — domestic, commercial *and* industrial — because breadth determines the ceiling and the license is the legal protection.
- **Confirm the physical side is one you can live with for decades.** You're in it now — pay attention to what the body feels like after a hard week, and to whether the February conditions are something you can sustain, because that's the thing most likely to end a career early.
- **Get fluent with the business and diagnostic tools as a professional, not a shortcut.** Learn the quoting, scheduling and invoicing software that's automating the office burden, and use thermal imaging and fault-finding instruments to inform judgment rather than replace it.
- **Feed curiosity and adaptability on purpose, and plan for ownership from the start.** The premium specializations — data centers, EV, solar and battery, heat pumps — shift over a career, and the income ceiling runs through running a business. The tradesperson who re-tools fastest and builds toward ownership compounds.

During the job hunt — aim for the durable work, don't just apply

- **Target the protected lanes on purpose.** Service, maintenance and retrofit field work, ideally electrical, ideally in high-demand specialisms — not whichever placement appears first.
- **Read the setting, not just the offer.** Repetitive production-line, prefab and modular work is the 5.0 rung — the controlled setting automation moves fastest into. If you take one, take it as a deliberate stepping-stone with an exit in mind, not as the plan.
- **Don't grab a job just because it's on offer in an exposed lane.** A role being available doesn't make it safe. An employer still running a large repetitive line or prefab intake may simply be behind the curve — and that's exactly the setting the score says drifts fastest. Availability can be a lagging indicator, not a green light.
- **Prefer breadth and the license.** A seat that gives you commercial and industrial work and counts toward the license beats a narrow domestic-only role that caps the ceiling, even if the second pays a little more today.
- **Ask the employer where AI and prefabrication are heading in the work.** Whether they're moving toward offsite modular assembly or staying in field service and retrofit tells you whether the seat holds its protection or quietly drifts toward the 5.0 end.
- **Weigh the trajectory and the body over the starting wage.** A role that builds toward ownership, breadth and a high-demand specialism beats a higher-paid seat in repetitive work that's harder on the body and closer to the automating end.

Go deeper — further reading

- [**Electricians — Occupational Outlook Handbook**](#) — US Bureau of Labor Statistics. Primary source and free: 9% projected growth through 2034, roughly 81,000 annual openings, wage distributions and entry requirements. Look up plumbing and HVAC while you're there.
- [**Electrician career guide 2026**](#) — the data center story in detail, including Microsoft's identification of the electrician shortage as the primary constraint on expansion, and the industry estimate of 300,000-plus additional electricians needed.
- [**Skilled trades statistics for 2026**](#) — wages, injury rates, union compensation differentials and the BLS opening projections collected in one place.

The counterargument — read this one:

The strongest case against our score isn't that robots will reach crawlspaces. It's **prefabrication**.

Modular and offsite construction is growing steadily, and it works precisely by moving work from unpredictable sites into controlled factories — the setting where our own framework says automation performs best. On this reading, the trades' protection erodes not because the work becomes automatable, but because the industry relocates it somewhere it already is.

Our own table concedes the mechanism: production-line and prefab work scores 5.0 against 2.5 for field service. If the share of construction done offsite rises substantially, the average tradesperson's exposure rises with it — and our headline figure would be describing a shrinking share of the trade.

Where we land: we hold the score because service, maintenance and retrofit work — which is a large and growing share of trade employment, driven by an ageing building stock — cannot be prefabricated. You cannot factory-build a repair to a 1963 house. But we'd treat rising offsite construction share as the single most important thing to watch in this guide, and we'll report it if it moves.

Bottom line

For the right person this is the strongest position in the index — and you're already on it. That's the whole reason being already in it is not a problem: the decisive moves, of setting and of specialism, were always going to happen from here.

The scores are among the lowest we measure. There is no tuition debt. The on-ramp is widening while most degree routes narrow. Demand is severe, well-documented, and — in electrical work — being driven by the AI buildout itself.

The protection is specific and worth understanding rather than assuming. It is not that the work is manual. It is that **the environment is different every time**, which is the condition automation handles worst. Move the same skills into a factory and the score doubles — which is why standing on the production-line rung is the one thing not to do.

The honest costs are real and have nothing to do with technology. Physical wear over decades is the thing most likely to end a career early. The income ceiling without business ownership is genuine. And construction is cyclical in a way that healthcare and education are not.

So the useful advice isn't a verdict on the trade — it's a direction. Steer toward service and retrofit work rather than repetitive new-build, chase breadth and the license rather than a narrow role, lean into the high-demand specialisms, and plan for ownership from the start rather than discovering at 35 that the ceiling is lower than expected. And build real fluency with the business and diagnostic tools now, inside whatever role you're in, because that's what converts a protected skill into a real ceiling.

And if anyone around you treated this as a lesser choice — it's worth knowing where that came from. On the evidence in front of us, this is a better financial and occupational position than most degrees currently offer, and you're already standing on it.

Questions worth sitting with

Whether you're the one in it or a parent thinking it through, these are the conversations that move the needle now. Read "you" as whoever the decision is about.

Where you actually stand

1. Which setting are you pointed at — and does it point at the *unpredictable field* (service, maintenance, retrofit) or the *controlled factory* (production line, prefab, modular)? 2. Looking at

the score table, which destination are you currently pointed at, and is it the one you'd choose on purpose? 3. Does your current path lead cleanly to the license and to breadth of work — domestic, commercial and industrial — or is it narrowing you toward one kind of job?

The move from here

4. What's the single most protected, highest-ceiling work you could realistically aim for from where you are now — electrical service, a data center specialism, retrofit — and what would make you a credible candidate for it? 5. If you're still training: are you getting the breadth of work and the clean licensing pathway you need, and is there more you could push for before it's no longer free? 6. If you're job-hunting: are you aiming at field service on purpose, or applying to whatever's open? And how honestly have you weighed the physical toll over decades against the version of the work you're choosing?

The AI question, practically

7. Of the five capabilities — run the business side, use diagnostic tools without deferring, look things up, specialize, get to ownership — which one could you visibly build in the next few months? 8. The income ceiling runs through running a trade business, and the office side is exactly what's automating. Can you point to a step you've taken toward ownership, or the nearest one to start with?

Testing it against reality

9. **The most valuable single action:** find someone ten years into the trade — ideally running their own business — and ask what their body feels like, whether they'd do it again, and how they got to the top end. 10. If the first offer that comes is a repetitive production-line or prefab role, what's your plan — take it as a deliberate stepping-stone with an exit in mind, or hold out for field service? Decide before the offer, not after.

This is analysis and informed judgment about a fast-moving situation, not a guarantee or a prediction. For the trades specifically, we've flagged that the most credible long-term threat is prefabrication moving work into controlled settings, rather than robotics reaching uncontrolled ones.

Scores for 2023 and 2025 are retrospectively calculated using the current methodology. Scores measure exposure to what AI can already do — not how much any particular employer has deployed.

Technical scoring appendix — Skilled Trades

Pivotum Degree Risk Index — Fall 2026 Edition

This appendix is the audit trail. It sets out the formula, the rating anchors, every factor rating behind every track score in this profile, the backcast arithmetic, a sensitivity analysis, and the specific things that would change our mind.

We publish it because a score nobody can check is an opinion with a decimal point.

1. The formula

$$\begin{aligned}\text{Risk} = & 0.35 \times \text{Automatability} \\ & + 0.15 \times \text{EntryErosion} \\ & + 0.15 \times (10 - \text{Physical}) \\ & + 0.15 \times (10 - \text{Trust}) \\ & + 0.10 \times (10 - \text{Regulatory}) \\ & + 0.10 \times (10 - \text{Judgment})\end{aligned}$$

Two exposure factors carry 50% of the weight; three protection factors carry 40%; judgment under novelty carries the remaining 10% and sits on the protection side. The deliberate consequence is that protection can outweigh exposure — which is why a highly automatable job like teaching still scores low.

Bands. Very low below 3.0 · Low 3.0–4.4 · Moderate 4.5–5.4 · Moderate–High 5.5–6.9 · High 7.0 and above.

2. Rating anchors

HOW THE 0–10 RAW RATINGS ARE ANCHORED

Every factor is rated on its own 0–10 scale before weighting. The anchors are identical across all twenty-eight careers, which is what makes the scores comparable.

Task automatability (A) — *what share of the working week could a capable current system already do to an acceptable standard?*

RATING	ANCHOR
0–2	Almost nothing. The work is physical, relational or wholly novel.
3–4	Administrative surround only — notes, scheduling, correspondence.
5–6	A meaningful minority of the week, mostly documentation and lookup.
7–8	A large share, including tasks central to the junior version of the role.
9–10	Most of the described duties, at or near professional standard.

Entry-path erosion (E) — *how much has the traditional route into this work narrowed?*

RATING	ANCHOR
0–2	Protected by statute or physical necessity. Training hours cannot be automated.
3–4	Stable. Some junior tasks absorbed; the apprenticeship survives.
5–6	Visible narrowing. Employers prefer experience; junior intake reduced.
7–8	Substantial. The work juniors learned on has largely gone.
9–10	The training layer and the automatable layer were the same layer.

Physical / embodied (P) — *must this be done in person, with a body, in an unpredictable setting?*

RATING	ANCHOR
0–2	Fully screen-based. Location-independent.
3–4	Occasional presence; the core work is not physical.
5–6	Regular physical presence, in largely controlled conditions.
7–8	Substantially physical, with some variability of setting.
9–10	Irreducibly physical, in conditions that differ every time.

Human trust / accountability (T) — *does someone need a specific human they can rely on, and must a human answer when it goes wrong?*

RATING	ANCHOR
0–2	Output is anonymous and reviewed by others.
3–4	Work is signed off by someone else. No personal reliance.
5–6	Some client relationship; accountability is institutional.
7–8	Named responsibility, with commercial or professional consequence.
9–10	The relationship is the service, and liability attaches personally.

Regulatory / licensing (R) — *does the law reserve this act to a qualified human?*

RATING	ANCHOR
0–2	No license exists and none is proposed.
3–4	Certification is customary but not reserved.
5–6	Partial reservation, varying by jurisdiction or task.
7–8	Licensed practice, with some unreserved adjacent work.
9–10	Comprehensive statutory reservation, including of the training route.

Judgment under novelty (J) — *how often does the work present something with no matching precedent, where being wrong is costly?*

RATING	ANCHOR
0–2	Routine by design. Deviation is an error, not a case.
3–4	Occasional exceptions, handled by escalation.
5–6	Regular non-standard cases within a known range.
7–8	Frequent genuine novelty with material consequences.
9–10	Novelty is constant, and in some fields adversarially generated.

3. Factor ratings by track

Ratings are the Fall 2026 edition unless a range is shown. **P, T, R and J are held constant across editions** — those protections are structural and do not move on a three-year horizon. All measured movement is in A and E.

ELECTRICIAN — SERVICE & MAINTENANCE — 2.5 (VERY LOW)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	3.9	4.4	4.9	1.72
Entry-path erosion	15%	1.5	1.8	2.0	0.3
Physical / embodied	15%	9.5	9.5	9.5	0.07
Human trust / accountability	15%	8.5	8.5	8.5	0.22
Regulatory / licensing	10%	9.0	9.0	9.0	0.1
Judgment under novelty	10%	9.0	9.0	9.0	0.1
				Total	2.52 → 2.5

PLUMBING / HVAC — SERVICE — 2.6 (VERY LOW)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	4.1	4.5	5.0	1.75
Entry-path erosion	15%	1.5	1.8	2.0	0.3
Physical / embodied	15%	9.5	9.5	9.5	0.07
Human trust / accountability	15%	8.5	8.5	8.5	0.22
Regulatory / licensing	10%	8.8	8.8	8.8	0.12
Judgment under novelty	10%	8.8	8.8	8.8	0.12
				Total	2.59 → 2.6

WELDING & FABRICATION — SITE WORK — 3.2 (LOW)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	4.5	5.0	5.4	1.89
Entry-path erosion	15%	2.0	2.2	2.5	0.38
Physical / embodied	15%	9.2	9.2	9.2	0.12
Human trust / accountability	15%	7.5	7.5	7.5	0.38
Regulatory / licensing	10%	7.5	7.5	7.5	0.25
Judgment under novelty	10%	8.2	8.2	8.2	0.18
				Total	3.19 → 3.2

CONSTRUCTION SITE TRADES — 3.4 (LOW)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	4.3	5.0	5.7	1.99
Entry-path erosion	15%	2.0	2.4	2.8	0.42
Physical / embodied	15%	9.2	9.2	9.2	0.12
Human trust / accountability	15%	7.5	7.5	7.5	0.38
Regulatory / licensing	10%	7.0	7.0	7.0	0.3
Judgment under novelty	10%	8.0	8.0	8.0	0.2
				Total	3.41 → 3.4

PRODUCTION LINE / PREFAB / MODULAR — 5.0 (MODERATE)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	3.9	5.1	6.4	2.24
Entry-path erosion	15%	3.5	4.0	4.5	0.67
Physical / embodied	15%	8.0	8.0	8.0	0.3
Human trust / accountability	15%	5.0	5.0	5.0	0.75
Regulatory / licensing	10%	4.5	4.5	4.5	0.55
Judgment under novelty	10%	5.0	5.0	5.0	0.5
				Total	5.01 → 5.0

4. Backcast arithmetic

Worked example for the headline track, showing exactly how the three-year movement is composed.

Electrician — service & maintenance

EDITION	A	E	PROTECTION DEFICIT (FIXED)	SCORE
2023	3.9	1.5	0.5	2.1
2025	4.4	1.8	0.5	2.3
Fall 2026	4.9	2.0	0.5	2.5

Movement decomposition: automatability contributed **+0.35** and entry-path erosion **+0.07**, for a total of **+0.4** points over three years.

The 2023 and 2025 figures are retrospective reconstructions using the current methodology, not archived scores from published editions. They are our best assessment of what each factor would have rated at the time, given what was then demonstrable. They are not measurements.

5. Sensitivity

How far the score moves if a single factor is rated one point differently:

FACTOR	±1 POINT MOVES THE SCORE BY
Task automatability	±0.35
Entry-path erosion	±0.15
Physical / embodied	±0.15
Human trust / accountability	±0.15
Regulatory / licensing	±0.10
Judgment under novelty	±0.10

What this means in practice. A one-point disagreement about automatability moves a score by 0.35 — enough to shift a track between bands at the margins. A one-point disagreement about licensing moves it by 0.10, which almost never changes a band.

So if you disagree with one of our numbers, the automatability rating is where the disagreement matters most, and it is the rating we would most want challenged.

6. Stated limitations

We measure capability, not deployment. Scores reflect exposure to what AI can already do at the leading edge — not how much any particular employer has adopted. Adoption lags capability by years and lags unevenly: large, well-funded organisations first; small, rural, public-sector and thin-margin employers much later. The lag is a buffer, not a shelter. It buys time without changing direction.

We do not measure demand. A task can be highly automatable and highly in demand simultaneously. Where the labor market and our framework disagree, we say so in the profile rather than adjusting the score to fit.

We do not measure oversupply. The number of people entering a field is outside our six factors, and for some careers it matters more than automation.

We do not measure industry economics. A profession can contract for reasons entirely unrelated to technology.

We do not measure whether the work is worth doing. Pay, satisfaction, debt, hours and meaning sit outside the score and are covered separately in each profile — often at greater length, because for several careers they matter more.

Sub-track definitions are ours. Job titles vary between employers and countries. We have chosen tracks that reflect genuinely different work, not official classifications.

7. What would change our mind

Each edition names specific, checkable indicators. If these move, the scores move — including downward.

Across the index:

- **Graduate hiring share.** If employers in the professions where we rate entry-path erosion highly return to hiring juniors at pre-2023 rates, those ratings fall.
- **Content of mandated training hours.** Several professions protect their on-ramp through required supervised experience. If the content of those hours shifts from producing work to supervising it, the protection weakens without any rule changing.
- **Offsite and prefabricated construction share.** The clearest test of our physical-protection ratings: automation performs well in controlled settings, so moving work indoors raises exposure without any robotics breakthrough.
- **Verified deployment rather than capability.** If adoption stalls materially below capability for several editions, our leading-indicator approach is overstating near-term risk and we will say so.

We will report movement in either direction. A framework that only ever revises upward is not measuring anything.

Scores measure exposure to what AI can already do — not how much any particular employer has deployed. The 2023 and 2025 figures are retrospective reconstructions using the current methodology.